Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/14/2020
PI Name:	Motil, Brian J. Ph.D.		
Project Title:	Modeling and Analysis of Packed Bed Reactor Experimental Data from International Space Station		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	03/01/2017	**End Date:**	06/24/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 6/24/2020; work will continue under new grant "Analysis of Packed Bed Reactor-2 Experimental Data from International Space Station" ; internal project and CoInvestigator grant 80NSSC20K0830 (Ed., 9/14/20) NOTE: Continuation per NSSC information of CoI's NNX14AD28G and NNX10AL37G to 80NSSC18K0813 with new title "Modeling and Analysis of Packed Bed Reactor Experimental Data from International Space Station" and new period of performance (Ed., 5/15/18, 4/28/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Balakotaiah , Vemuri Ph.D. (University of Houston (grants 80NSSC18K0813, NNX17AF91G))		
Grant/Contract No.:	Internal Project ; CoI grants 80NSSC18K0813, NNX17AF91G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Modeling and Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity" internal project--Principal Investigator Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant NNX14AD28G (and previously NNX10AL37G) to grant 80NSSC18K0813 with new title "Modeling and Analysis of Packed Bed Reactor Experimental Data from International Space Station" and new period of performance. Includes also information for project “GAS-LIQUID TWO-PHASE FLOW THROUGH PACKED BEDS IN MICROGRAVITY,” internal project--Principal Investigator Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant NNX17AF91G (Ed., 5/15/18, 4/28/21) A basic understanding of two-phase flows through porous media is of interest in many chemical and biological processing systems as well as many geophysical applications. NASA technologies include adiabatic two-phase flows or two-phase flows with heat transfer that are relevant to life support systems, fuel cells, in-situ resource utilization, heat pipes, materials processing, production of pharmaceutical grade water, and transport of nutrients in soils. A very important unit operation that involves gas-liquid flow through porous media is the fixed Packed Bed Reactor (PBR). Examples of current space systems in which a fixed PBR is used include the Volatile Removal Assembly (VRA), the Integrated Advanced Water Recovery System (AWRS), and IntraVenous Water GENeration system (IVGEN). NASA is currently operating systems with packed bed reactors for ground and flight-based testing with an incomplete understanding of how the reduced gravity environment affects the performance and reliability of these reactors. This is especially critical when gas-liquid flows are involved. The expected outcome of this research effort is to develop a set of guidelines and tools to enable engineers to reliably design and operate packed bed reactors for microgravity as well as the lunar and Martian environments. Existing approaches for predicting the flow regimes and pressure drop for gas-liquid flow through a PBR are valid only for 1 g, and are based on liquid (or gas) holdup within the bed. Initial microgravity tests conducted in the reduced gravity aircraft and drop tower clearly show that the liquid holdup is significantly influenced by gravity. Furthermore, most of these models have only been verified using the "trickle flow" regime which does not exist in microgravity. Basic hydrodynamic models developed specifically for reduced gravity have been partially validated within the constraints of aircraft and ground facilities. However, typical flow ranges used for real systems require up to several minutes to develop fully which is far beyond the time scales available for ground tests. In addition, instabilities and hysteresis effects that may exist require long testing durations to establish new operating conditions. Thus, testing on the International Space Station (ISS) is required for complete validation of the semi-empirical models. Flight experiments can provide critical hydrodynamic information for a project with a broader scope that includes reduced gravity aircraft and ground-based (1 g) experiments. The main objective of the overall project is to develop and validate macroscopic equations that can be used in partial and microgravity conditions to accurately predict flow pattern transitions; pressure drops; and mass transfer rates in chemical and biological reactors in which gas-liquid flows occur through randomly packed beds. The hydrodynamic investigations addressed in the Packed Bed Reactor Experiment (PBRE) focus on the transitions between flow regimes (e.g., bubbly-to-pulse flow transition) and the associated pressure gradients over the range of relevant operating regimes of the PBRs (e.g., liquid and gas flow rates, particle sizes, and fluid viscosities). These design tools provide a fundamental understanding of these phenomena in low gravity. [Ed. Note (September 2020): Project continues as "Analysis of Packed Bed Reactor-2 Experimental Data from International Space Station," with Dr. Brian Motil continuing as the Principal Investigator (PI); Dr. Vemuri Balakotaiah continues as CoInvestigator with grant 80NSSC20K0830. Period of performance is 6/25/2020-12/4/2021]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The PBRE provides a fundamental investigation of two phase flow through porous media in the absence of gravity. This enables a direct study of the other forces that impact the flow and reaction rates such as surface tension, inertia, and viscous forces. These new insights will impact reactor designs on Earth.
Task Progress:	The main goal of this work was to analyze the data obtained in the Packed-Bed Reactor Experiment (PBRE) conducted aboard the International Space Station (ISS). The specific objectives of the PBRE-ISS project were to develop and validate macroscopic equations that can be used in microgravity conditions to accurately predict the pressure drop and flow patterns (and transition between flow patterns) in chemical and biological reactors in which gas-liquid flows occur through randomly packed beds. The PBRE-ISS experimental data was analyzed to determine the pressure drop in bubbly flow with a focus on the viscous-capillary regime, which is the main regime of operation of most reactors in reduced gravity. The pressure drop data from PBRE-ISS experiments was also compared with data obtained in ground (aircraft) based experiments. Investigator Balakotaiah and University of Houston (UH) researchers (Dr. Paul Salgi, Ms. Mahsa Taghavi) collaborated with Drs. Motil and Rame of NASA Glenn Research Center (GRC) in analyzing and organizing the data on the pressure drop as well as flow patterns. The results of the analysis were recently summarized in a refereed journal article (See Bibliography section) and are expected to be published in the near future. In addition to analyzing the data from PBRE-ISS experiments, all researchers at UH also worked with Drs. Motil and Rame to identify changes to the experimental protocol for the re-flight PBRE-2 experiments which are planned to be conducted in the summer of 2020. [Ed. Note (September 2020): Project continues as "Analysis of Packed Bed Reactor-2 Experimental Data from International Space Station," with Dr. Brian Motil continuing as the PI; Dr. Vemuri Balakotaiah continues as CoInvestigator with grant 80NSSC20K0830. Period of performance is 6/25/2020-12/4/2021]
Bibliography Type:	Description: (Last Updated: 04/26/2024)
Articles in Peer-reviewed Journals	Motil BJ, Rame E, Salgi P, Taghavi M, Balakotaiah V. "Gas-liquid flows through porous media in microgravity: The International Space Station Packed Bed Reactor Experiment." AlChE J. Early view online 11 September 2020:e17031. https://doi.org/10.1002/aic.17031 , Sep-2020
Articles in Peer-reviewed Journals	Taghavi M, Balakotaiah V. "Gas hold-up and bubble behavior in an upflow packed bed column in the limit of low flow rate." AlChE J. 2019 Aug;65(8):e16624. https://doi.org/10.1002/aic.16624 , Aug-2019

NASA Technical Documents	Balasubramaniam R, Rame E, Motil BJ. "Microgravity Liquid-Gas Two-Phase Flow: Review of Pressure Drop and Heat Transfer Correlations and Guidelines for Equipment Operability." Cleveland, Ohio : NASA Glenn Research Center, 2019. NASA/TM-2019-220147. https://ntrs.nasa.gov/api/citations/20190001795/downloads/20190001795.pdf , Mar-2019